41. Domain of $f(x) = \sqrt{x+5}$

- A square root is defined for non-negative numbers.

$$x + 5 \geq 0 \implies x \geq -5$$

✅ **Answer: b) x ≥ –5**

---

42. Domain of $f(x) = \frac{1}{x^2-9}$

- Denominator ≠ 0:

$$x^2 - 9 \neq 0 \implies x \neq \pm 3$$

✅ **Answer: a) All x except ±3**

---

43. Which function is one-one?

- One-one: each output corresponds to exactly one input.
- x^2 → not one-one ($x^2 = (-x)^2$)
- $|x|$ → not one-one
- x^3 → one-one (monotonic)
  ✅ **Answer: c) x^3**

---

44. Which function is onto R?

- Onto: range = codomain.
- x^2 → [0,∞), not R
- e^x → (0,∞), not R
- x^3 → (-∞,∞), yes onto R
  ✅ **Answer: c) x^3**

---

45. A function is invertible if and only if it is:

- Invertible → must be **bijective** (one-one + onto).
  ✅ **Answer: d) Bijective**

---

46. Range of $f(x) = |x|$

- Absolute value is always ≥ 0 → [0, ∞)
  ✅ **Answer: c) [0,∞)**

---

47. Domain of $\log(x-2)$

$$x-2>0 \implies x>2$$

✅ Answer: b) x > 2

---

48. Domain of $\sqrt{4-x}$

$$4-x \geq 0 \implies x \leq 4$$

✅ Answer: a) x ≤ 4

---

49. Range of $y=\frac{1}{x^2+1}$

- $x^2+1 \geq 1 \implies 1/(x^2+1) \leq 1$, >0
- So range: (0,1]
  ✅ **Answer: a) (0,1]**

---

50. $f(x)=x^2$ is invertible on:

- Not one-one on all R.
- Restrict to x ≥ 0 → one-one
  ✅ **Answer: b) x ≥ 0**

---

51. $f: R \to R, f(x)=x+5$

- Linear, one-one and onto, so bijective
  ✅ **Answer: d) All**

---

52. $f(x)=\sin x$ is invertible on:

- Sin is one-one on [-π/2, π/2]
  ✅ **Answer: c) [−π/2, π/2]**

---

53. $f(x)=\cos x$ is invertible on:

- Cos is one-one on [0, π]
  ✅ **Answer: b) [0,π]**

---

54. Which is NOT a function?

- Circle $x^2 + y^2 = 4$ → fails vertical line test
  ✅ **Answer: a) Circle $x^2+y^2=4$**

---

55. Relation is a subset of:

- Relation: set of ordered pairs
  ✅ **Answer: c) A×B**

---

56. $f(x) = 5$ is:

- Constant function
  ✅ **Answer: b) Constant**

---

57. Domain of $\tan x$

- Undefined at x = odd multiples of π/2
  ✅ **Answer: b) All real except odd multiples of π/2**

---

58. Range of $\tan x$

- Tan covers all real numbers
  ✅ **Answer: b) All real**

---

59. Range of $\cos x$

- Cos varies from -1 to 1
  ✅ **Answer: b) [−1,1]**

---

60. Range of $\sin x$

- Sin varies from -1 to 1
  ✅ **Answer: c) [−1,1]**

---

61. A function f:A→B must satisfy:

- Every element of domain has exactly one image
  ✅ **Answer: a) Every a∈A has an image**

---

62. Many-one function means:

- Several inputs can give same output
  ✅ **Answer: b) Several different inputs give same output**

---

63. A function f:X→Y is onto if:

- Range = codomain
  ✅ **Answer: b) Range = Codomain**

---

64. Number of functions from A (3 elements) to B (4 elements)

- Each element of A → 4 choices → $4^3 = 64$
  ✅ **Answer: a) 64**

---

65. Cardinality of A×A if |A|=4

$$|A \times A| = 4 \times 4 = 16$$

✅ **Answer: c) 16**

---

66. If f and g are one-one, then g∘f is:

- Composition of one-one functions → one-one
  ✅ **Answer: b) One-one**

---

67. Inverse exists if f is:

- Must be bijective
  ✅ **Answer: c) Bijective**

---

68. Range of secx

$$\sec x = 1/\cos x$$

- Cos ∈ [-1,1] → sec ∈ (-∞,-1] ∪ [1, ∞)
  ✅ **Answer: b) (−∞,−1] ∪ [1,∞)**

---

69. $f(x) = x^3$ is:

- Monotonically increasing
  ✅ **Answer: a) Increasing**

---

70. $y = \sqrt{x^2}$

- Always non-negative → |x|
  ✅ **Answer: c) |x|**

---

71. Range of e^x

- Always positive → (0, ∞)
  ✅ **Answer: a) (0,∞)**

---

72. $\log x$ inverse is:

- Inverse of log → exponential
  ✅ **Answer: a) e^x**

---

73. Relation "is father of" is:

- Anti-symmetric (cannot have a "father" in both directions)
  ✅ **Answer: c) Anti-symmetric**

---

74. Relation "perpendicular to" is:

- Symmetric: if A ⊥ B, then B ⊥ A
  ✅ **Answer: c) Symmetric**

---

75. Identity functions are:

- One-one and onto → bijective
  ✅ **Answer: d) All**

---

76. $f(x) = |x - 3|$ ≥

- Absolute value ≥ 0
  ✅ **Answer: a) 0**

---

77. Domain of $f(x) = 1/|x|$

- Denominator ≠ 0 → x ≠ 0
  ✅ **Answer: a) x≠0**

---

78. Domain of $\sqrt{x^2 - 4}$

$$x^2 - 4 \geq 0 \implies x \geq 2 \text{ or } x \leq -2$$

✅ **Answer: b) x≤−2 or x≥2**

---

79. y = $\sin^{-1}x$ is defined for:

- Inverse sine defined for |x| ≤ 1
  ✅ **Answer: c) |x| ≤ 1**

---

80. Range of $\cos^{-1}x$

- Cos^{-1} x ∈ [0, π]
  ✅ **Answer: b) [0,π]**

---

81. Range of $\tan^{-1}x$

- $\text{Tan}^{-1}x$ ∈ (-π/2, π/2)
  ✅ **Answer: a) (−π/2, π/2)**

---

82. If f(x) = x^3, f(−x) =

- Odd function → f(−x) = −f(x)
  ✅ **Answer: b) −f(x)**

---

83. Reflexive relation must contain:

- (a,a) for all a ∈ A
  ✅ **Answer: c) (a,a) for all a∈A**

---

84. x divides y is:

- Reflexive, transitive, antisymmetric → partial order
  ✅ **Answer: d) All**

---

85. Parity relation (“same parity”)

- Equivalence relation (reflexive, symmetric, transitive)
  ✅ **Answer: c) Equivalence**

---

86. f:R→R, $f(x) = x^5$ is:

- Odd-degree polynomial → bijective
  ✅ **Answer: d) All**

---

87. $f(x)=x^2$ not invertible because:

- Many-one
  ✅ **Answer: b) Many-one**

---

88. f(x) = sin2x is:

- Periodic, period π
  ✅ **Answer: a) Periodic**

---

89. Composition is:

- Not always commutative
  ✅ **Answer: c) Sometimes**

---

90. Range of 1/x

- Cannot be 0 → all real except 0
  ✅ **Answer: b) All real except 0**

---

91. If f(x)=0 for all x, it is:

- Constant
  ✅ **Answer: b) Constant**

---

92. tanx has period:

- π
  ✅ **Answer: a) π**

---

93. sinx has period:

- 2π
  ✅ **Answer: b) 2π**

---

94. If f is invertible, $f \circ f^{-1}$ is:

- Identity function
  ✅ **Answer: b) Identity function**

---

95. Range of |x−2|:

- Always ≥ 0 → [0, ∞)
  ✅ **Answer: b) [0, ∞)**

---

96. Composite of constant and non-constant:

- Constant (constant function applied to anything is constant)
  ✅ **Answer: a) Constant**

---

97. Domain of √(2x−1):

$$2x - 1 \geq 0 \implies x \geq 1/2$$

✅ **Answer: c) x≥1/2**

---

98. Range of $f(x)=x^2+4$:

- Minimum at x=0 → f(0)=4 → [4, ∞)
  ✅ **Answer: b) [4,∞)**

---

99. Range of |x|:

- [0, ∞)
  ✅ **Answer: b) [0,∞)**

---

100. Domain of cotx:

- Undefined where sinx=0 → x≠nπ
  ✅ **Answer: a) x≠nπ**

101. $f(x) = 0/x$ is undefined at:

- $x=0$
  ✅ Answer: a) $x=0$

---

102. $f(x)=\sec x$ undefined at:

- $\cos x=0 \rightarrow x = \pi/2 + n\pi \rightarrow \pi/2$
  ✅ Answer: a) $\pi/2$

---

103. $f(x)=\csc x$ undefined at:

- $\sin x=0 \rightarrow x = n\pi$
  ✅ Answer: d) $n\pi$

---

104. $f(x)=|x|$ is:

- Even function
  ✅ Answer: a) Even

---

105. $f(x)=x^3$ is:

- Odd function
  ✅ Answer: b) Odd

---

106. $|x|$ has minimum at:

- $x=0$
  ✅ Answer: a) $x=0$

---

107. Graph of odd function symmetric about:

- Origin
  ✅ Answer: c) Origin

---

108. Graph of even function symmetric about:

- y-axis
  ✅ Answer: b) y-axis

109. Range of $\tan^2 x$:

- Square of real numbers → [0, ∞)
  ✅ **Answer: a) [0, ∞)**

---

110. Domain of $f(x)=1/(x+3)$:

- Denominator ≠ 0 → $x \neq -3$
  ✅ **Answer: b) $x \neq -3$**

---

111. Number of one-one functions from A={1,2,3} to B={a,b,c}?

- Permutations → 3! = 6
  ✅ **Answer: a) 6**

---

112. Number of relations on set with 4 elements

- Relations = subsets of A×A → 2^16 = 65536
  ✅ **Answer: d) 65536**

---

113. Ceiling function gives:

- Smallest integer ≥ x
  ✅ **Answer: b) Smallest integer ≥ x**

---

114. $f(x)=e^x$ is:

- One-one, onto (0, ∞), invertible
  ✅ **Answer: d) All**

---

115. sinx not invertible because:

- Not one-one on R
  ✅ **Answer: b) Not one-one**

---

116. cosx not invertible because:

- Not one-one on R
  ✅ **Answer: b) Not one-one**

117. tanx is invertible on:

- (-π/2, π/2)
  ✅ **Answer: a) (−π/2, π/2)**

118. 1/x is:

- Odd function → 1/(-x) = -1/x
  ✅ **Answer: b) Odd**

119. |x−3| ≥:

- Always ≥ 0
  ✅ **Answer: a) 0**

120. f(x)=x^5 is:

- Odd-degree polynomial → bijective
  ✅ **Answer: a) Bijective**